

Democratic Economic Cell Network

From the national foundation to the planetary horizon —
Patch v5.0.3: the Good Life Index replaces GDP as the
reference indicator.

Version 5.0.3 — Good Life Index

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v5.0.3 makes GDP an internal variable and adopts the Good Life Index (GLI) — the geometric mean of a “social foundation × ecological ceiling” — as its headline indicator, more faithful to the model's sufficiency principles.

Patch application note v5.0.3

PATCH V5.0.3 · 26 JUNE 2026

Objective: replacing the headline indicator. GDP measures monetary volume, not quality of life or respect for boundaries — it even rewards the overproduction the model seeks to abolish. v5.0.3 makes GDP an **internal variable** (Sovereign Fund calculation, income elasticity) and adopts the **Good Life Index (GLI)** as the reference indicator. Key result: the Reference Scenario reaches **GLI = 83.5/100**; a higher-growth variant (GDP 385 instead of 280) falls back to 72.2 because it degrades material sufficiency. Measured by the GLI, the model's sober scenario is superior — which GDP masked.

PATCH V5.0.2 · 26 JUNE 2026

Objective: extension to the 9 planetary boundaries (food & plastics). This patch corrects the blind spot of v5.0.1: the exclusive focus on energy decarbonisation. An energy system, even 100 % renewable, addresses only two of the seven planetary boundaries transgressed in 2025 (climate and ocean acidification). To guarantee a habitable planet, the model now structurally integrates food and materials management.

The general architecture (Layers 0 to 17) is maintained. The change concerns **Layer 12 (Ecological loop)**, whose strict “carbon budget” is replaced by the global framework of the **9 planetary boundaries**, grounded in the *Planetary Health Check 2025* and *EAT-Lancet 2.0*. Two new protocols complete the layer.

This note summarises the changes; the versioning, formula and layout fixes of v5.0.1 remain fully valid (allocation constraint F1 written $\alpha + \beta + \gamma + \delta + \varepsilon = 1$, ε threshold set at 2 % of GDP, energy balance F4, terminology of the Commons Usage Levy and the Commons Guard).

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1. Executive summary

RCED proposes a radical yet pragmatic transformation of the Swiss economic system. v5.0.2 preserves the entirety of the foundation's achievements (v4.3.2 then Layers 13-14 of v4.4.1) — blockchain transparency, cooperative governance, democratic control, shared infrastructure, ecological loop and Decarbonisation Fund — applies patch v4.4.3 (use-ownership), then patch v5.0.2 (9 planetary boundaries) and finally patch v5.0.3, which makes GDP an internal variable and adopts the **Good Life Index (GLI)** as the reference indicator.

Patch **v4.4.3** rests on four pillars: (1) capitalist private property gives way to use-ownership; (2) classic redistributive taxation is replaced by a Commons Usage Levy; (3) the model shifts from a mere fiscal-transition tool to a direct social-production structure; (4) the defence of the commons is rigorously structured.

Patch **v5.0.2** adds two protocols to Layer 12: **food reform** (agroecology + sharp reduction in meat, protocol 12.A) and the **limitation of novel entities** (plastics regulation, protocol 12.B). Together, energy + food + materials cover the seven planetary boundaries transgressed today.

Key results — Reference Scenario (Year 20)

The Reference Scenario projections (Part I) demonstrate exceptional macroeconomic and ecological performance without money creation, while maintaining a strict gains-allocation constraint.

83.5 / 100 Good Life Index Doughnut reference (soc. × eco.)	72 / 100 Gross National Happiness +14-point increase	0.185 Gini coefficient Strong reduction (−0.135)
28.5 h Working hours / week Time freed (−12.5 h)	−68 % Net emissions Net-zero 2050 trajectory (LCA)	19.7 % Sovereign Fund surplus Expressed as % of GDP

1.1 From the GDP index to the Good Life Index

GDP measures neither quality of life nor respect for planetary boundaries, and rewards the overproduction that use-ownership precisely seeks to abolish. v5.0.3 demotes it to an internal variable and replaces it with the **Good Life Index (GLI)**, inspired by “doughnut” economics: the geometric mean of a *social foundation* and an *ecological ceiling*, so that neither can offset the collapse of the other.

Definition
GLI = $\sqrt{F_{\text{soc}} \times F_{\text{eco}}}$ — where F_{soc} aggregates well-being (GNH), equality (1–Gini), time autonomy and needs met; F_{eco} aggregates the climate trajectory, planetary boundaries respected and material sufficiency. Each half-index is itself the geometric mean of sub-scores normalised to 100.

Sub-index (/100)	Baseline 2026	Reference (Y20)	High-growth variant
Well-being (GNH)	58	72	100
Equality (1–Gini)	68	81.5	82
Time autonomy	29	81	74
Needs met	55	90	90
→ Social foundation	50.2	80.9	85.9
Climate trajectory	0	85	89.6
Planetary boundaries	28.6	100	100
Material sufficiency	0	75.3	25
→ Ecological ceiling	0.0	86.2	60.7
Good Life Index	≈ 0	83.5	72.2

STRUCTURAL RESULT · V5.0.3

The higher-growth variant (GDP 385 instead of 280) achieves a higher GDP but a **lower GLI** (72.2 vs 83.5): its extra output is bought at the cost of lower energy sufficiency (energy 92.5 vs 77.4), which collapses the ecological ceiling from 86.2 to 60.7. Since the geometric mean forbids any compensation, those GDP points translate into a *loss* of good life. GDP rewarded material excess; the GLI penalises it — confirming the sufficiency vs green-growth argument structurally (§ 5.3).

Complementary indicators

Indicator	Y20 value	Impact / Justification
Pooled networks	3 networks	Rail, energy and digital open and rent-free.
Final energy (Y20 index)	77.4	22.6 % drop in final consumption.
Required overbuild (Layer 14)	1.25×	Optimised vs ~2× required without a broad interconnected grid.
Planetary boundaries covered (v5.0.2)	7 / 7	Energy + food + materials address the 7 transgressed boundaries.
Health co-benefit (12.A)	↑ ↑	Up to 15 M deaths avoided/yr (healthy diet).
GDP (index, now internal)	280.4	+180.4 % over 20 years; feeds the Sovereign Fund calculation, no longer a success metric.

2. Evolution from v4.3.2 to v5.0.3

Version v4.4.x invalidates none of the v4.3.2 corrections but extends them; v5.0.2 carries this logic toward non-energy boundaries. The foundation (Layers 0 to 12) and the eight numerical-consistency checks are unchanged. The model is anchored in an incremental logic relying on existing precedents (SBB, Swissgrid, civilian service, EU energy communities) and on foundations drawn from Marx (property), Ostrom (commons, graduated sanctions) and radical subsidiarity.

Element (base)	Evolution with the patch	Justification / Impact
γ reallocation (non-automatable sector)	Layer 13: qualifying citizen service (renovation, energy, care).	Provides a concrete operator for the 3–5× pace required for a 100 % renewable system.
Decarbonisation Fund (δ)	Layer 13: thermal storage (sand) + local resources.	Removes dependence on critical minerals; seasonal storage; lowers final consumption.
National commons (Layer 11)	Layer 14: federated energy cells, energy subsidiarity.	Smooths intermittency across space, cuts overbuild from ~2× → 1.25×.
Strict carbon budget (Layer 12)	v5.0.2: framework of the 9 planetary boundaries + protocols 12.A (food) and 12.B (plastics).	Addresses the 5 boundaries unreachable by the energy transition alone (biosphere, land, freshwater, N/P flows) and chemistry/plastics.
Headline indicator: GDP	v5.0.3: Good Life Index (GLI), GDP becoming an internal variable.	Measures quality of life under the ecological ceiling rather than monetary volume; aligns evaluation with the model's sufficiency.
International electricity-scale limit	Part II: Earth Federation Project, sequenced after foundation maturity.	Treats the political blind spot as a horizon distinct from the deliverable layers.
Capitalist private property & redistributive taxation	Institutionalised use-ownership and Commons Usage Levy (Layer 2).	The good is tied to its actual use. No more after-the-fact redistribution; billing at source.
Financial-wealth register (Layer 1)	Usage cadastre (energy, land, digital).	Auditing targets the real use of resources, consistent with use-ownership.
Collective security (§14, horizon)	Commons Guard + Layers 15-17 (defence, immunity, verification).	Graduated defence of the commons: enclosure prevention, automatic immunity, human validation.

3. Paradigm shift (patch v4.4.3)

Purpose: the model no longer presents itself as a mere fiscal-transition tool, but as a direct social-production structure: economic coordination runs through the use of the commons rather than through accumulation and late redistribution.

3.1 From private property to use-ownership

The patch abolishes private property as defined by capitalism (the capital title as a right to exclude and extract rent). It substitutes **use-ownership**: a good is constitutionally tied to whoever actually uses it according to the rules of the commons, not to a detachable title. Inspired by Marx reread through Ostrom, this principle does not forbid owning what one uses (housing, tools, a farm), but removes the ability to capture rent on an unused good that deprives others of its use.

3.2 From redistributive taxation to the commons usage levy

The redistributive tax system (Piketty / Zucman / Saez) is entirely replaced by a **Commons Usage Levy**. Instead of levying ex-post on incomes and wealth to redistribute, the model bills at source the use of common resources (energy, land, data, register capacity). The contribution becomes a condition of access to the commons, in the Ostromian logic of access-dues. Layer 0 (inequality diagnosis) remains relevant as a starting point, but the instrument shifts to the usage levy.

4. The full layer stack (foundation + patch)

The overall structure integrates the historical foundation layers, reinterpreted by patch v4.4.3, complemented by the ecological-transition innovations (extended 12, 13-14) and security ones (15-17).

#	Element / Layer	Institutional function & mechanism
0	Diagnosis	Inequality assessment (analytical starting point, no longer an action mechanism).
1	Usage register	Cadastre of energy, land, digital and carbon usage on Blockchain.
2	Usage levy	Replaces redistributive taxation: billing commons use at source.
3	Capitalisation	Citizen sovereign fund investing directly in the real economy.
4a	Production	Worker cooperatives.
4b	Consumption	Consumer cooperatives and democratic supply planning.
5	Health / Time	Solidarity insurance and a reduced working week.
6	Coordination	Federated cells, using the free market as a mere transmission signal.
7	Governance	Swiss model based on strict subsidiarity, from municipalities to the Confederation.
8	Knowledge	Official training (SERI / OrTra) with credentialed reconversion schemes.
9	Propagation	0 % investment enabling cells to spread and form internationally.
10	Time	Labour indexation: required time falls in proportion to automation.
11	National commons	Rail, energy and digital in open access to purge the economy of rents.
12	Ecological loop (extended v5.0.2)	Framework of the 9 planetary boundaries (Planetary Health Check 2025 / EAT-Lancet 2.0). Includes the carbon budget + Decarbonisation Fund (δ), food reform (12.A) and regulation of novel entities / plastics (12.B).
13	Citizen mobilisation	Citizen service + sober thermal storage + structural relocalisation.
14	Energy cells	Nested energy sharing and networked energy subsidiarity.
S.I.	Commons Guard	Internal security: enclosure prevention, technical audit, administrative interposition.
15	Territorial defence	Subsidiary, decentralised external defence, funded by the commons dividend.
16	Immunity	“Broadcast” protocol: automatic, reversible isolation of offending cells.
17	Verification	Decentralised media consensus (human-in-the-loop) to validate genuine aggression.

4.1 Gains-sharing mechanism (recap)

For each year t , the annual automation rate $\Delta P(t) = r$ (estimated between 2 % and 6 %) is rigorously allocated according to the following constraint:

ALLOCATION CONSTRAINT

$$\alpha + \beta + \gamma + \delta (+ \varepsilon) = 1$$

The operating foundation's default coefficients:

- $\alpha = 45 \%$ → feeds working-time reduction ($\alpha \times \Delta P(t)$).
- $\beta = 35 \%$ → feeds the citizen Sovereign Fund ($\beta \times \Delta P(t)$).
- $\gamma = 10 \%$ → sectoral reallocation and training / Layer 13 ($\gamma \times \Delta P(t)$).
- $\delta = 10 \%$ → Decarbonisation Fund / Layer 12 ($\delta \times \Delta P(t)$).
- ε (federal contribution / Part II) → conditional, activated only if the Sovereign Fund surplus exceeds 2 % of GDP.

Golden rule & democratic governance

100 % of automation gains stay allocated, with no money creation whatsoever. The annual arbitration and the modulation of the coefficients (α , β , γ , δ , ε) follow no fixed algorithm: the allocation is sovereignly deliberated and validated by a Citizens' Assembly, aligning production and working-time reduction with real needs.

5. Extended Layer 12 — The 9 planetary boundaries

NEW · PATCH V5.0.2

The strict “carbon budget” of v5.0.1 is replaced by the global framework of the 9 planetary boundaries (Planetary Health Check 2025, EAT-Lancet 2.0). A 100 % renewable system addresses only 2 of the 7 boundaries transgressed in 2025 (climate, ocean acidification). Two protocols cover the rest.

5.1 Protocol 12.A — Food reform

Transition to **agroecology** combined with a **sharp reduction in meat consumption**. Food is the leading driver of boundary transgression; acting on it unlocks five boundaries unreachable by the energy transition alone.

- **Boundaries addressed:** biosphere integrity, land-system use, freshwater, and biogeochemical flows (nitrogen N / phosphorus P).
- **Land synergy:** lower meat consumption is the prerequisite that frees farmland. It makes agroecology — whose per-area yield is lower — compatible with the land-use boundary.
- **Health co-benefit:** up to 15 million deaths avoided per year, which massively strengthens the political acceptability of the measure.

5.2 Protocol 12.B — Novel entities (plastics)

Limiting plastic use through **regulation**: targeted bans, deposit-return schemes, and extended producer responsibility (REACH-type EPR). This protocol addresses the 7th transgressed boundary — chemistry / novel entities — which is moreover **aggravated by the material demand inherent to the energy transition**.

Lever (v5.0.2)	Targeted planetary boundaries	Mechanism
Energy foundation (Layers 12-14)	Climate · Ocean acidification	100 % renewable, sufficiency, meshed grid.
12.A Food reform	Biosphere · Land · Freshwater · N/P flows	Agroecology + meat reduction (land release).
12.B Novel entities	Chemistry / plastics	Bans, deposit-return, EPR (REACH-type).

5.3 Keeping the RCED paradigm — sufficiency vs green growth

A “social-democratic” variant (keeping private property + public investment + regulation) also reaches the viability zone on the 7 transgressed boundaries. The radical architecture of full RCED is nonetheless retained for durability against material growth.

Limit of conventional regulation

The social-democratic model patches each boundary with policies grafted onto an intact growth engine, making the risky bet of absolute material decoupling.

RCED robustness

RCED holds the zone by its very structure. Use-ownership removes the incentive to rent and therefore to overproduction: it structurally removes the material engine, ensuring a more robust standing within planetary boundaries over the long term.

6. Layer 13 — Citizen mobilisation of the transition

The deployment pace required to substitute renewables for fossil fuels is 3 to 5 times the current rate. The conventional labour market can only supply this workforce at a problematically slow pace. Layer 13 articulates three levers to accelerate the transition while lowering final energy consumption without degrading well-being:

- **Citizen transition service** — repurposing an existing institution toward critical worksites (§ 6.1).
- **Sober thermal storage** — sand batteries for seasonal heat storage (§ 6.2).
- **Renovation & relocation** — structural reduction of final energy demand (§ 6.3).

6.1 Citizen transition service

The principle is to repurpose a consensual, existing Swiss institution (compulsory military/civilian service) rather than build unprecedented coercion. Fit young adults perform a qualifying service, supervised by professionals, channelled to the transition's critical worksites: building-stock energy renovation, installation of renewable infrastructure and heat pumps, ecological restoration and support to the care sector.

Mandatory safeguards to embed

- **Individual freedom:** the obligation remains a constraint; it is justified if the service stays short, qualifying and guarantees a real choice of track.
- **Productivity:** an untrained conscript is less effective than a skilled tradesperson. The service must complement skilled trades, not replace them.
- **Accounting:** the opportunity cost must be booked as a labour transfer (not as free labour).

6.2 Sober thermal storage — sand batteries

A fundamental technical distinction: you don't store electricity, you store heat. Sand is raised to 500–600 °C using surplus renewable electricity, then releases that heat over days or months via urban district-heating networks.

What sand storage solves	What must be soberly acknowledged
Mineral independence: no lithium, cobalt, nickel or rare earths. Sand, steel and a heating element — a local, abundant, neutral resource.	Poor electrical efficiency: heavy losses when reconvertng to electricity. Unbeatable for heating, unsuited to mobility.
Seasonal storage: shifts energy from summer to winter — which electrochemical storage cannot do at scale.	Heavy infrastructure: assumes vast heat networks, an infrastructure project in itself.
Peak management: absorbs overproduction peaks instead of curtailing them.	Complementarity: an indispensable complement — not a substitute — to electrochemical storage (mobility, electronics).

6.3 Renovation, relocation and energy balance

In a temperate/cold-climate country, building heating accounts for nearly 45 % of final energy consumption. A deep thermal retrofit cuts heat demand by 50 to 80 %, while a heat pump (COP 3 to 4) drastically divides the primary energy required.

Transition item	Measured effect on final consumption
Building heating (retrofit + HP)	↓↓↓ Dominant, priority lever.
Freight transport (relocalisation)	↓ Net reduction of physical flows.
Seasonal storage (sand)	Neutral — secures the 100 % renewable thermal supply.
Specific electricity (overall electrification)	→ or ↑ Stable to slightly rising consumption.
Embodied energy of the transition (worksites)	↑ Temporary rise, amortised thereafter.
Net balance (Y20 index = 77.4)	Overall 22.6 % drop in final consumption, at constant well-being.

7. Layer 14 — Federated energy cells

The residual blind spot of the transition concerns the specific intermittency of electricity. Layer 14 answers it through networked sharing, in synergy with storage. The model articulates two physical dimensions: **time** (moving energy from one moment to another via local storage) and **space** (moving energy from one geographic point to another via the meshed grid).

Scale	What spatial sharing solves	What it cannot solve alone
Neighbourhood	Fine demand diversity, lower line losses, optimised self-consumption.	Day/night cycle, local weather hazards (zone production stays correlated).
National	Smoothing of microclimates and regional weather differences.	Day/night cycle (Switzerland's longitudinal span is too narrow).
Continental	Strong weather compensation, exploiting time-zone offsets and wind/solar anti-correlation.	Deep, persistent winter-night lulls (Dunkelflaute).
Global	Full resolution of the day/night cycle (it is always daytime on one face of the Earth).	Physically robust, but reopens the immense challenge of political governance (Part II).

7.3 Guiding principle: energy subsidiarity

The grid operates on a fractal, nested architecture: local self-sufficiency is maximised first. Each higher tier intervenes only to regulate the residue the lower tier cannot absorb. Territorial autonomy is the inviolable rule, global interconnection the safety net — never the reverse. You do not depend on the large grid; you are connected to it as insurance.

Impact on oversizing

Thanks to the broad grid and Layer 14's meshing, the overbuild index collapses from ~2x (a stand-alone island system) to just **1.25x**, massively saving embodied energy and critical minerals. Layer 1 (Blockchain) traces energy flows and peer-to-peer reciprocity credits. Real precedents: ZEV/RCP self-consumption groupings in Switzerland and EU citizen energy communities.

Earth Federation Project

A multigenerational project, not deliverable within 20 years. Its activation remains strictly conditional on the full maturity of Part I (network propagation to a critical mass of countries).

8. Status, sequencing and nature of the problem

The Earth Federation differs fundamentally from Layers 13 and 14: those solved technical locks through engineering; the Earth Federation claims to fill the political blind spot — global coordination. Yet sovereignty does not vanish through a mere design trick. The two projects are strictly sequenced: each phase must prove its robustness before the next.

10. Equalisation through the commons dividend

To be politically tractable, global equalisation is not organised as a direct transfer of wealth between continents, but rests on a dividend from the planetary commons. The atmosphere (single carbon register), the international energy backbone, Earth orbit, the oceans and critical-mineral deposits are defined as global commons. Levies collected at source feed an Earth-scale Sovereign Fund: the paradigm is no longer “the rich pay for the poor” but “the planetary commons pays its dividend and distributes it according to need”.

11. Financing: 0 % investment → commons → dividend

The global financial engine reuses Layer 9's propagation mechanism, inspired by the great cooperative federations (Migros, Coop, Raiffeisen, Mondragon) where members' surpluses finance the apex structures. National cells' surplus is channelled, via 0 % lending, toward infrastructure (interconnection cables, registers, environmental restoration); once operational, these generate access levies financing structural redistribution.

Collecting without an army — access-dues

Following Ostrom, the global contribution is not an invasive customs or military levy, but an unavoidable condition of access to the commons network. To take part in trade, use the energy infrastructure or the carbon register, a cell must be current on its access-dues. You do not coerce by force: you disconnect technical access.

14. Collective security: from non-violent leverage to interposition corps

The most sensitive and most tightly framed component of Part II. Designed as a last resort, it excludes any centralised standing army and rests on two graduated instruments:

Instrument	Nature & mechanism	Resort
Non-violent leverage	Exclusion from commons access (carbon register, unified market, energy interconnection). No wealth is extracted; the cell is technically disconnected.	First resort. Soft and economical.
Consented interposition corps	Physical deployment between belligerents to freeze hostilities. Strict blue-helmet logic, never assault troops.	Last resort. Heavy and exceptional.

Absolute red lines

- **Interposition, not imposition:** the federal corps does not deploy against one side to subdue or occupy it. It does not govern and levies no tax.
- **Prior consent:** deployment strictly by invitation or consented mandate; an absolute ban on entering a non-consenting sovereign state by force.
- **Enforcement dilemma:** without coercion, the register is just a dashboard; endowed with enforcement power, it would drift toward an authoritarian world-state. The Ostromian confederal approach with enumerated competences serves as a firewall.

Annex A — Mathematical formulas

F1 — Extended macroeconomic allocation constraint$\alpha + \beta + \gamma + \delta + \epsilon = 1$ Where $\epsilon \geq 0$ is the federal contribution (Part II), capped and strictly zero if the Sovereign Fund surplus is $\leq 2\%$ of GDP.
F2 — Retrofitted building heat demand & HP consumption$Q_{heat}(t) = Q_{heat}(0) \times (1 - \tau_{retro})$$E_{elec} = Q_{heat} / COP$ Target parameters: $\tau_{retro} \approx 0.5$ to 0.8 $COP \approx 3$ to 4.
F3 — Seasonal thermal storage (sand)$S_{th}(t) = S_{th}(t-1) + \eta_{charge} \times E_{surplus}(t) - Q_{released}(t) - losses(t)$ Thermal-to-thermal cycle optimised for absorbing instantaneous renewable surpluses.
F4 — Cell energy balance (Layer 14)$Self-cons = \min(Prod_{local}, Cons_{local})$$Surplus_{shared}(t) = \sum \max(Prod - Cons, 0) \rightarrow deficit\ cells$ Oversizing target (overbuild): $1.25\times$ (vs $\sim 2\times$ without a meshed grid).
F5 — Cascading federal financing (Part II)$Invest_{commons}(t) = \epsilon \times \Delta P(t) \times GDP(t)$$Dividend_{commons}(t) = price_{carbon} \cdot E_{taxed} + royalties_{minerals} + levies$$Equalisation(t) = allocate(Dividend_{commons})$
F6 — Planetary viability index (new v5.0.2)$Viab(t) = \prod_{i=1..9} \mathbb{1}[Pressure_i(t) \leq Boundary_i]$ The system is viable only if the 9 planetary boundaries are simultaneously respected. Energy protocols cover $i \in \{\text{climate, oceans}\}$; 12.A covers $\{\text{biosphere, land, freshwater, N, P}\}$; 12.B covers $\{\text{novel entities / plastics}\}$.
F7 — Good Life Index (GLI) — new v5.0.3$GLI(t) = \sqrt{(F_{soc}(t) \times F_{eco}(t))}$$F_{soc} = (GNH \cdot (1-Gini) \cdot 100 \cdot Time \cdot Needs)^{1/4}$$F_{eco} = (Climate \cdot Viab \cdot Sufficiency)^{1/3}$ Sub-scores normalised to 100. The geometric mean forbids a strong dimension from offsetting a collapsed one: a high GLI requires both the social and the ecological. GDP no longer appears as an objective; it remains an internal variable (basis of the Sovereign Fund and income elasticity).

Annex B — Glossary (new entries)

Term	Definition and systemic role
Use-ownership	Legal framework tying a good to its actual user under the charters of the commons. Removes the validity of detachable capital titles enabling passive rent extraction (Marx reread through Ostrom).
Usage levy	Mechanism billing the use of common resources at source, fully replacing the former ex-post redistributive-tax schemes (Layer 2).
Commons Guard	Civil and administrative internal-security body: register audit (Layer 1), alerting the assemblies, administrative interpositions (access exclusions) to block enclosure.
Immunity (broadcast)	Automated technical protocol for the reversible, immediate isolation of a cell breaching the common rules. An instance of the Ostromian graduated sanction.
Verification	Decentralised media consensus mandatorily including human validation (human-in-the-loop) before any collective response.
Planetary boundaries (v5.0.2)	Framework of the 9 environmental boundaries (Planetary Health Check 2025) within which humanity can prosper sustainably. Replaces the strict carbon budget as the ecological loop (Layer 12).
Agroecology (v5.0.2)	Agricultural system of protocol 12.A, combined with a sharp meat reduction. Addresses biosphere, land, freshwater and N/P flows; the land release from lower meat ensures compatibility with the land-use boundary.
Novel entities (v5.0.2)	The “chemistry / plastics” boundary (protocol 12.B), regulated by bans, deposit-return and extended producer responsibility (REACH-type). Aggravated by the material demand of the energy transition.
Good Life Index (GLI) (v5.0.3)	Headline indicator replacing GDP. Geometric mean of the social foundation and the ecological ceiling (“doughnut” economics), on a 0–100 scale. Measures a good life held within boundaries rather than a volume of production.
Social foundation (v5.0.3)	Half-index of the GLI aggregating well-being (GNH), equality (1–Gini), time autonomy and essential needs met.
Ecological ceiling (v5.0.3)	Half-index of the GLI aggregating the climate trajectory, respect for planetary boundaries and material sufficiency (final energy). The model's limiting factor.
Interposition corps	International peacekeeping force (a thin core of professionals + a reserve of volunteers) positioned strictly between belligerents to block violence.
Citizens' Assembly	Sovereign deliberative body modulating and validating the annual allocation of the macroeconomic parameters (α , β , γ , δ , ϵ) outside any technocratic automatism.

Annex C — Sources and references

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- **Polar Night Energy (2022).** — First industrial high-temperature sand thermal-storage unit (Finland) — reference for Layer 13.
- **Cooperative models** (Mondragon, Migros, Raiffeisen). — Federal structures financing their development by pooling activity surpluses.
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